

NTX Morse BBuBIOL box, Giant Microbit, and Safe.

Parts list. Items one off unless stated.

BBuBIOL box.

Plastic base, 3D printed. Box, laser cut 4mm plywood.

Mechanical.

Machine screws A2 Stainless, M4 x 10mm, 4 off

Brass hot melt knurled insert M2.5 x 4mm, 4 off

Stick on rubber feet, 10mm dia x 4 off.

Electronic/ Electrical.

BBC Microbit version version 2, with 2xAA battery box.

Breakout board for Microbit, RK Education RKuB1, angle connector.

Prototype board RK education RKPt170

Power chassis socket to match 2A 5V psu.

2 or 3 pin plug and socket for LED strips.

4 way plug and socket terminal strip, 3 off

Phono plug and socket for audio connection

Push buttons, Red and Blue, 12mm hole.

Terminal strip 5A, 12 way, 2 off

Transistor 2N2222A, 5 off

8mm dia LED green, and yellow

RGBW SK6812 LED, single pixel cut from stirp, 2 off

Resistor 10K, 0.25W, 3 off

Resistor 1K, 0.25W, 2 off

Resistor 270R, 0.25W, 2 off

Dupont jumper wire 20cm for Microbit/ Arduino/ Rasp Pi, Male Male ends

Giant Microbit.

Back board, laser cut 5mm MDF. Front face laser cut 4mm Acrylic. Details film cut and adhesive back attached see drawings.

Mechanical.

Plastic ducting cover 25mm wide, 0.5m long, cut to make LED strip clamps

Stainless screws 3.5mm dia, 10mm, 8 off

Cable tidy adhesive bases, 5 off

Electronic/ Electrical.

Audio amp 10W, 5-12V supply, with volume control

Speaker 10W, 8 ohm

Terminal strip 5A, 4 way

Terminal strip 5A, 12 way

Transistor 2N2222A

RGBW SK6812 LED, 3 pixels pixel cut from strip, 25 off

8mm dia LED red 2 off

Resistor 270R, 0.25W, 2 off

Hookup wire as required

Safe.

Laser cut 4mm MDF, see drawings.

Mechanical.

Plastic ducting cover 25mm wide, 0.5m long, cut to make LED strip clamps

Stainless screws 3.5mm dia, 10mm, 10 off

30mm dia, 40mm long, coil spring, for door open assist

30mm hinges, 2 off

Electronic/ Electrical.

Flashing beacon, 12V, car lighter plug connection

Chassis socket for car lighter plug

Free socket for chosen 12V power adaptor

Solenoid lock mechanism, 5V

Servo, type MG90S

Fused terminal strip, for 20mm fuse

Terminal strip 5A, 12 way

Transistor 2N2222A, 2 off

Transistor TIP121, 2 off

RGBW SK6812 LED, 20 pixels pixel cut from strip, 2 off

RGBW SK6812 LED, 25 pixels pixel cut from strip, 2 off

Resistor 100K, 0.25W

Resistor 10K, 0.25W, 3 off

Resistor 1K, 0.25W, 3 off

Hookup wire as required